Elektrostatik

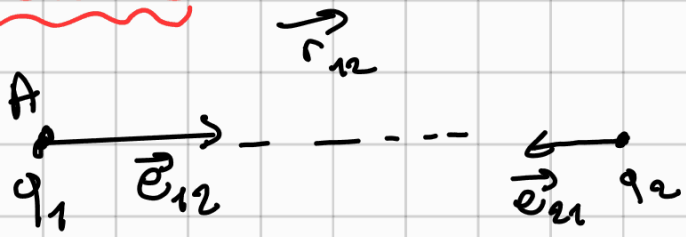
$$\vec{e}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$$

$$|\vec{e}_{12}| = |\vec{e}_{21}| = 1$$

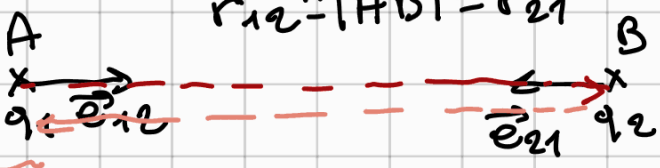
$$\vec{e}_{21} = -\vec{e}_{12}$$

$$\vec{r}_{12} = r_{12} \cdot \vec{e}_{12}$$

$$\vec{r}_{21} = r_{12}\vec{e}_{21} = -r_{12}\vec{e}_{12}$$



$$r_{12} = |\overrightarrow{AB}| = r_{21}$$



$\varepsilon_0 = \frac{1}{36\pi} 10^{-9}$ F/m
Boşluğun elektrik sabiti

$$\vec{r}_{12} = \overrightarrow{AB} = r_{12} \cdot \vec{e}_{12} = -\vec{r}_{21} = -r_{21}\vec{e}_{21}$$

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{(r_{12})^2} \vec{e}_{12} = -\vec{F}_{21}$$

q_1'in $q_2$ye uyguladığı kuvvet

q_2'nin q_1'e uyguladığı kuvvet



$$\vec{e}_{AB} = \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|}$$

$$\vec{e}_{BA} = -\vec{e}_{AB} = -\frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = \frac{\overrightarrow{BA}}{|\overrightarrow{AB}|}$$

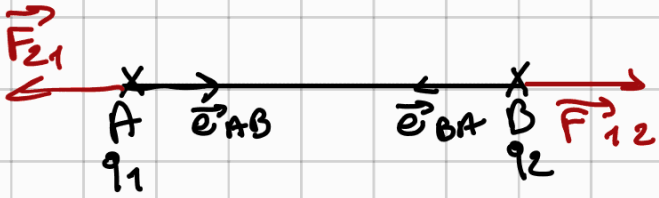
$q_1 \cdot q_2 > 0$

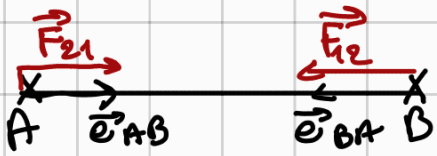
$q_1 > 0$ veya < 0

$q_2 > 0$ veya < 0

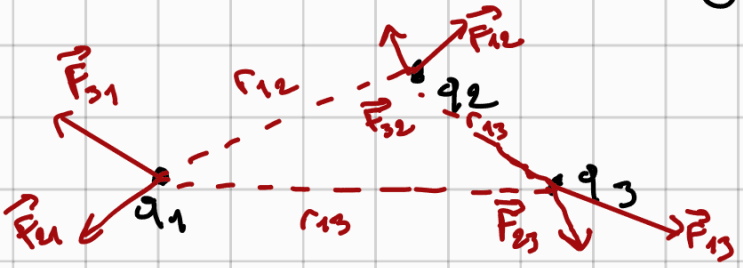
$q_1 \cdot q_2 > 0$ ise yükler birbirini iter

$q_1 \cdot q_2 < 0$ ise yükler birbirini çeker ($q_1+ \; q_2-$ / $q_2+ \; q_1-$)

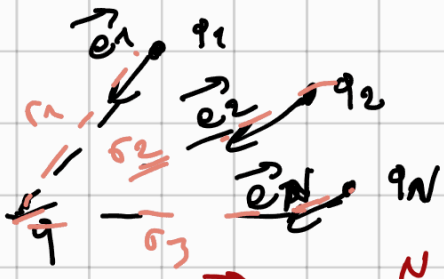




★İki denge yük halinde superposition ilkesi geçerlidir

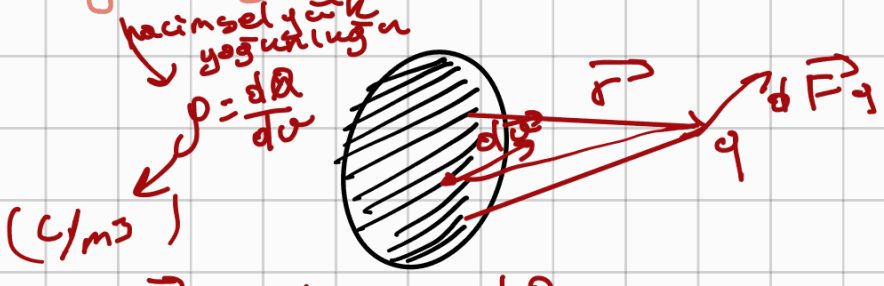


$\vec{F}_1 = \vec{F}_{31} + \vec{F}_{21}$ $\quad \vec{F}_2 = \vec{F}_{12} + \vec{F}_{32}$ $\quad \vec{F}_3 = \vec{F}_{23} + \vec{F}_{13}$



$$\vec{F}_q = \frac{q}{4\pi\varepsilon_0} \sum_{n=1}^{N} \frac{q_n}{(r_n)^2} \vec{e}_n$$

Eğer yükler ayrık değil, sürekli bir yoğunlukla dağılmışsa



$$d\vec{F}_q = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q\,dQ}{r^2} \rho \vec{} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q\rho\,dv}{r^2} \vec{e}$$

$9 \cdot 10^9 N$

$$\Rightarrow \vec{F}_q = \int_v d\vec{F}_q = \int_v \frac{1}{4\pi\varepsilon_0} \frac{q\rho\,dv}{r^2} \cdot \vec{e}$$

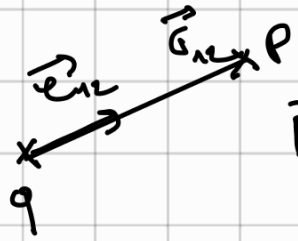
$$\vec{F}_q = \frac{q}{4\pi\varepsilon_0} \int_v \rho \frac{\vec{r}}{r^3} dv$$

$$\vec{F}_1 = \frac{1}{4\pi\varepsilon_0} q \int \frac{\rho}{r^3} \cdot \vec{r}\, dv$$

$$\rho = \rho_{iç} + \rho_{dış}$$

$$\vec{E}_{dış} \triangleq \vec{F}_q$$

Elektrostatik alan $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho \vec{r}}{r^3} dQ$ $\quad 10^{-15}_{m} < r < \infty$



$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{[r_{12}]^3}\vec{r}_{12}$$

$[\vec{E}] = V/m = N/C$ $\quad [q] = C$ $\quad [F] = N$ $\quad [\varepsilon_0] = F/m$

$[|\vec{r}|] = m$

Uygulama

Pr.-2. Bir proton ile elektron arasında oluşan mekanik ve elektrik çekim kuvvetlerinin oranını hesaplayınız. Aynı şeyi iki elektron için tekrarlayınız.

$\frac{2e}{3} = k_1$ $\quad \frac{-e}{3} = k_2$

$N \cdot e$ ← Maddesel ise

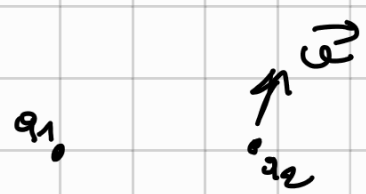
$N \in \mathbb{Z}$

$p = 2k_1 + k_2$

$n = 3k_1 + 2k_2$

★ eknot

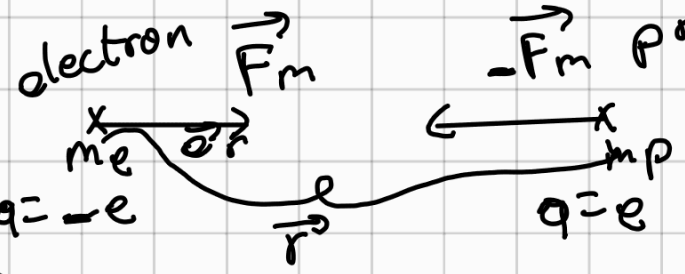
q_1 durgun ve q_2 hareketliyse coulomb yasası bir yönde geçerlidir. Durgunun hareketliye olanı geçerlidir.



$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r_{12}^3}\vec{r}_{12}$$

$$\vec{F}_{12} \neq -\vec{F}_{21}$$

$\vec{r} = r\vec{e}_r$



$\varepsilon_0 = \left(\frac{1}{36\pi}\right) \cdot 10^{-9}$ F/m

$m_e = 9,1 \cdot 10^{-31}$ kg

$m_p = m_n = 1,67 \cdot 10^{-27}$ kg

$e = 1,6 \cdot 10^{-19}$ C

$k = 6,67 \cdot 10^{-11} \frac{N m^2}{kg^2}$

$$\vec{F}_{mec} = k \cdot \frac{m_e \cdot m_p}{r^2} \cdot \vec{e}_r$$

$$\vec{F}_{elec} = \frac{1}{4\pi\varepsilon_0} \frac{|q_e||q_p|}{r^2} \cdot \vec{e}_r = \frac{1}{4\pi\varepsilon_0} \cdot \frac{e^2}{r^3} \vec{r}$$

$|\vec{F}_{mec}| = F_m$

$|\vec{F}_{elec}| = F_{elec}$

$$\frac{F_{mec}}{F_{elec}} = \frac{k \cdot m_e m_p \cancel{|\vec{r}|}}{\cancel{r^3}} = \frac{1}{\frac{1}{4\pi\varepsilon_0} \frac{e^2}{\cancel{r^3}} \cancel{|\vec{r}|}} = 4\pi\varepsilon_0 k \frac{m_e m_p}{e^2}$$

$$\Rightarrow F_{elec} = \frac{e^2}{m_e \cdot m_p} \cdot \frac{1}{4\pi\varepsilon_0 k} F_{mec}$$

$$F_{elec} = \frac{(1,6) \cdot 10^{-38} \cdot 1}{(9,1)(1,67) 10^{-31} \cdot 10^{-27}} \cdot \frac{10^{11}}{4\pi\varepsilon_0 6,67} \cdot F_{mec}$$

$$\approx \frac{10^{40}}{4} F_{mec} //$$